

# Resultate der Hyperparameter Analysis

---

**Date:** 2025-3-27

**Author:** Flo

**Links:** [MA-Besprechung 2025-03-27](#)

## Methodik

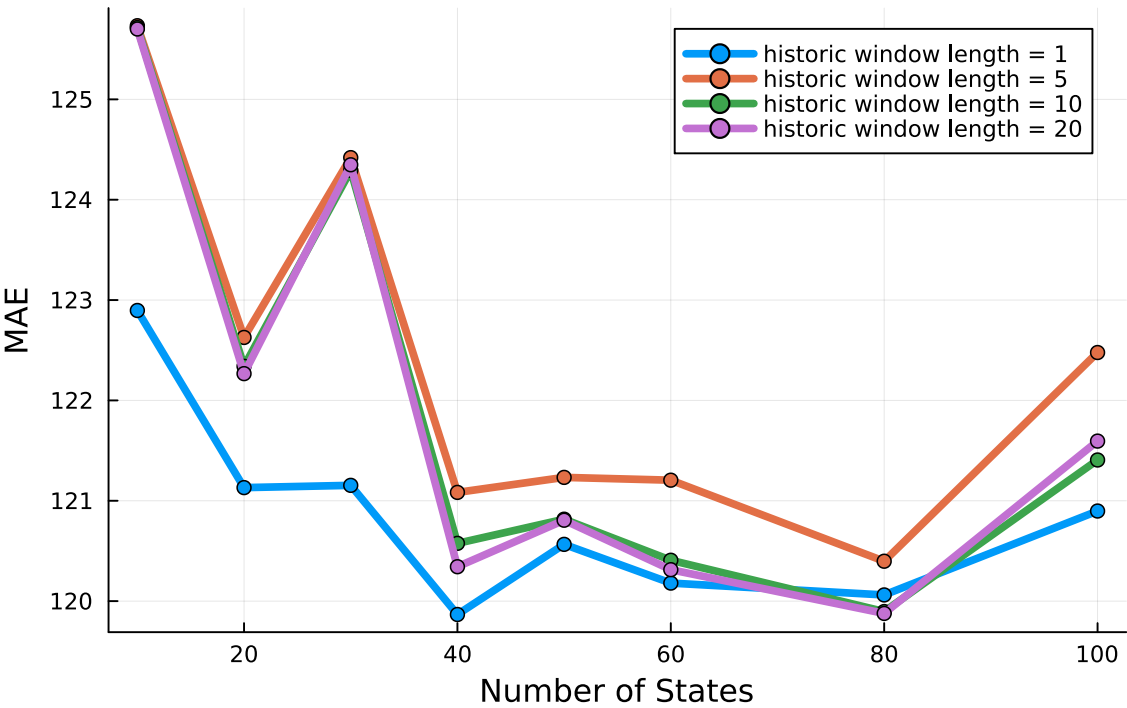
---

- Basismodel:
  - Eindimensionaler Dataset mit Stromverbrauch (Watt)
  - 1 Jahr Training
  - Hyperparameter: **Number of States**
- Vorhersage:
  - Erwartungswert der Verteilungsvorhersage (=Erwartungswertvorhersage)
  - One-Step-Ahead Sliding-Window Prediction:
    - Ein-Schritt-Prognose: Nur nächste Viertelstunde wird vorhergesagt
    - Sliding-Window: Länge der historischen bekannten Beobachtungen sind fixiert (= **historic window length**)
  - Sensitivitäts-Analyse: Test different **historic window length**
- Evaluation:
  - Eval-Dataset: ersten 2 Wochen jeder Season
    - Kommentar: Einschränkung auf nur 5 Evaluationen wie vorausgesetzt nicht aussagekräftig => dh kompletten Testdatensatz genommen
  - Variation von **Number of States** und **historic window length**
    - **Number of States** in {5, 10, 20, 30, 40, 50, 60, 80, 100}
    - **historic window length** in {1, 5, 10, 20}
  - MAE der Erwartungswertvorhersage
  - Residual Variance der Erwartungswertvorhersage

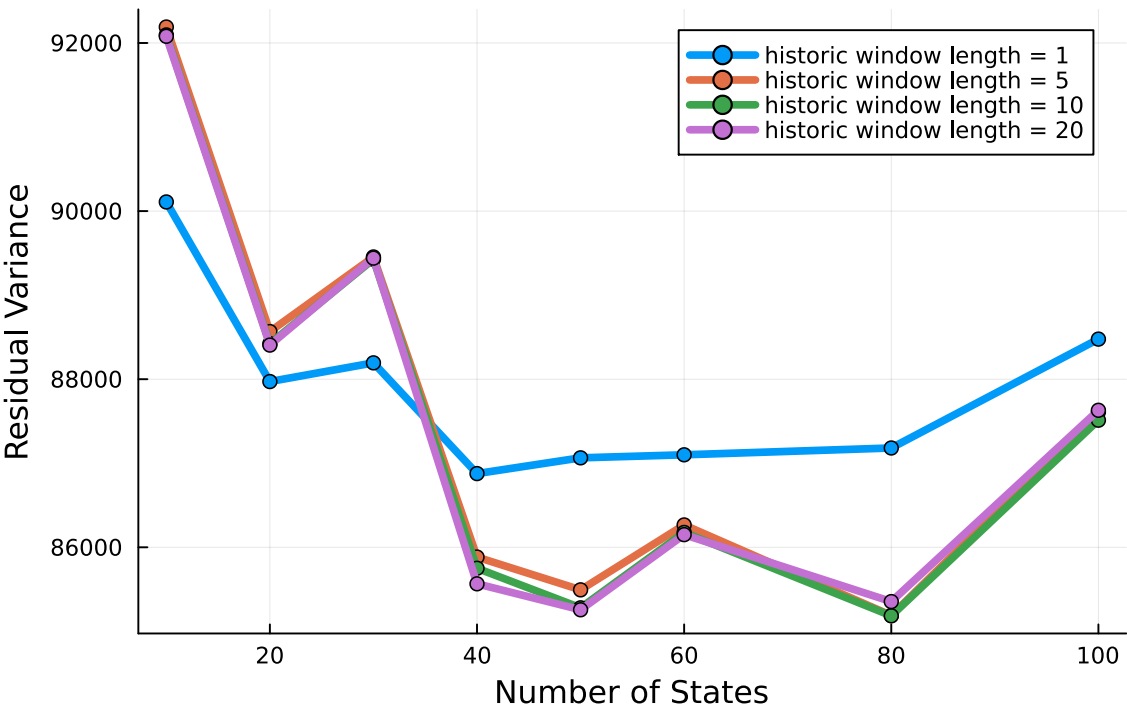
Results

- **Graphic Results** for HH 1 (= household number 1)

Hyperparameter Analysis: MAE



Hyperparameter Analysis: Variance of Residuals



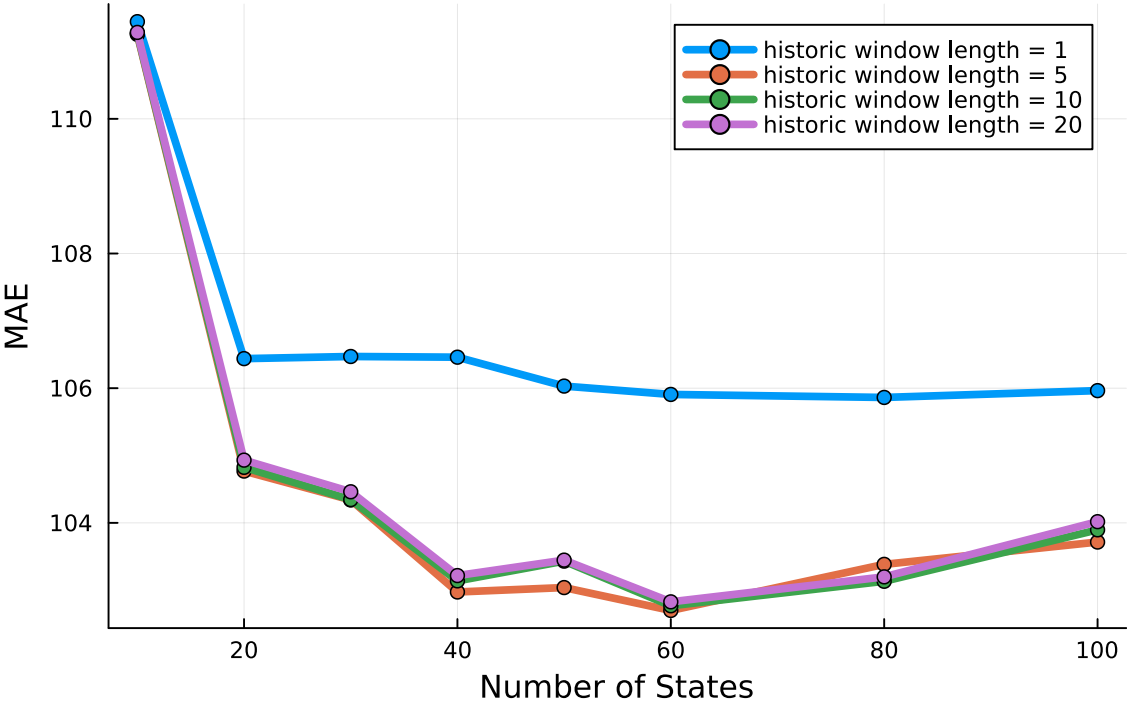
- **Numeric Results** for HH 1
  - MAE: Lowest score at **Number Of States** = 80 and **Historic Window Length** = 20

| MAE | 1       | 5       | 10      | 20      |
|-----|---------|---------|---------|---------|
| 10  | 122.896 | 125.734 | 125.716 | 125.701 |
| 20  | 121.131 | 122.628 | 122.340 | 122.267 |
| 30  | 121.154 | 124.419 | 124.293 | 124.348 |
| 40  | 119.867 | 121.083 | 120.576 | 120.343 |
| 50  | 120.566 | 121.233 | 120.816 | 120.806 |
| 60  | 120.180 | 121.205 | 120.407 | 120.313 |
| 80  | 120.062 | 120.398 | 119.896 | 119.878 |
| 100 | 120.898 | 122.477 | 121.406 | 121.595 |

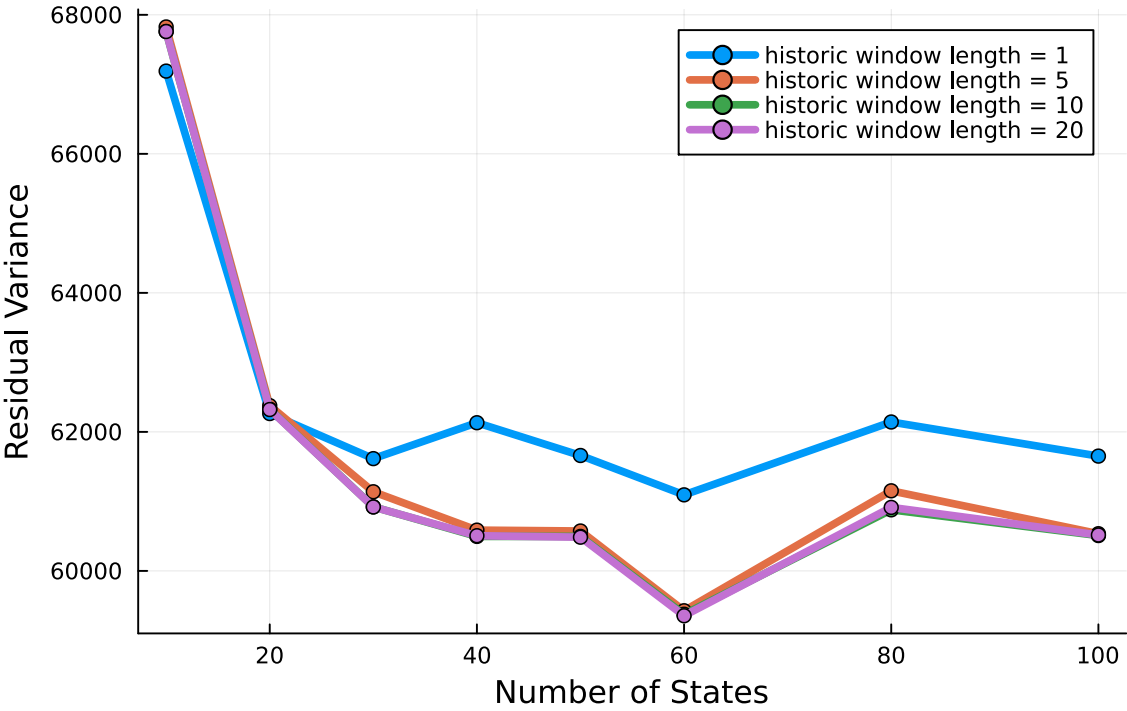
| Residual Variance | 1       | 5       | 10      | 20      |
|-------------------|---------|---------|---------|---------|
| 10                | 90108.7 | 92190.0 | 92095.9 | 92079.8 |
| 20                | 87971.8 | 88566.5 | 88422.9 | 88405.9 |
| 30                | 88194.4 | 89454.1 | 89427.2 | 89440.3 |
| 40                | 86877.0 | 85883.7 | 85750.5 | 85565.6 |
| 50                | 87064.4 | 85493.8 | 85279.9 | 85256.7 |
| 60                | 87100.4 | 86265.9 | 86176.8 | 86150.5 |
| 80                | 87181.5 | 85188.8 | 85184.9 | 85354.0 |
| 100               | 88477.0 | 87609.9 | 87512.3 | 87630.2 |

- **Graphic Results** for HH 2 (= household number 2)

Hyperparameter Analysis: MAE

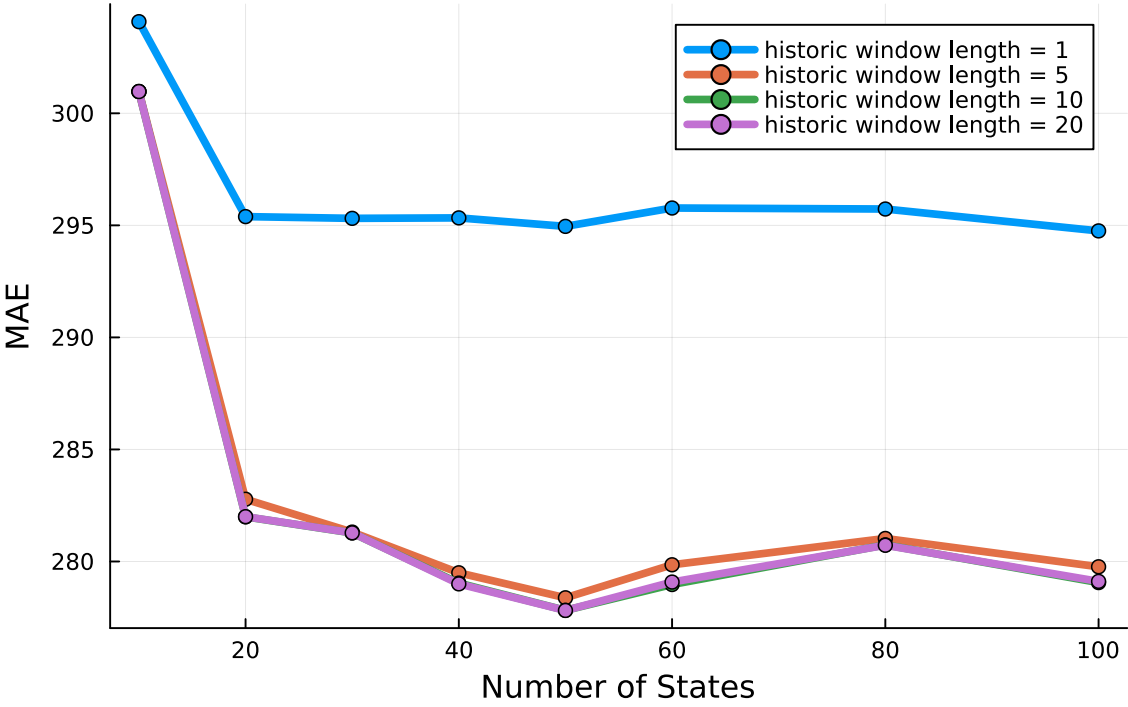


Hyperparameter Analysis: Variance of Residuals



- **Graphic Results** for HH 3 (= household number 3)

Hyperparameter Analysis: MAE



Hyperparameter Analysis: Variance of Residuals

